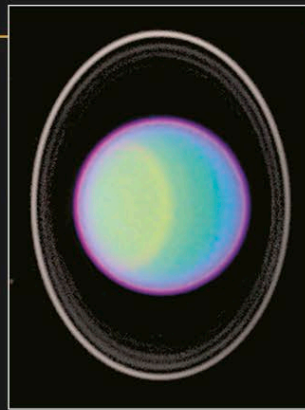


THE GAS GIANTS

The four large planets immediately beyond the Asteroid Belt are called the gas giants. These planets have many properties in common. Each has a core composed of rock and ice. This is surrounded by a liquid or semisolid mantle containing hydrogen and helium, or—in the case of Uranus and Neptune—a combination of methane, ammonia, and water ices. Each has a deep, often stormy atmosphere composed mainly of hydrogen and helium. All four have a significant magnetic field, but Jupiter's is exceptional, being 20,000 times stronger than that of Earth. Each of the gas giants is orbited by a large number of moons, several dozen in the case of Jupiter. Finally, all four gas giants have ring systems made of grains of rock or ice. These rings may have been present since the planets formed, or they may be the fragmented remains of moons that were broken up by the gas giants' powerful gravitational fields.



URANUS AND RINGS

Uranus has 11 major rings and a blue coloration caused by the presence of methane in its atmosphere. (This is a Hubble infrared image.) Its spin axis is tilted over on its side.

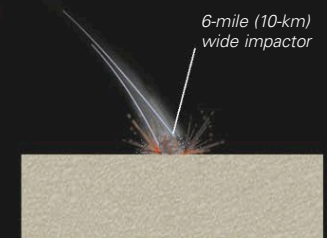
THE ROCKY PLANETS

The four inner planets of the solar system are also called the rocky planets. They are much smaller than the gas giants, have few or no moons, and have no rings. All four were born in a molten state due to the heat of the collisions that led to their formation. While molten, the materials from which they are made became separated into a metallic core and a rocky mantle and crust. Throughout their later history, all these planets suffered heavy bombardment by meteorites that left craters on their surfaces, although on Earth these craters have largely become hidden by various geological processes. In some other respects, the rocky planets are quite diverse. For example, Venus has a dense atmosphere consisting mainly of carbon dioxide, while Mars has a thin atmosphere composed of the same gas. In contrast, Mercury has virtually no atmosphere and Earth's is rich in nitrogen and oxygen.

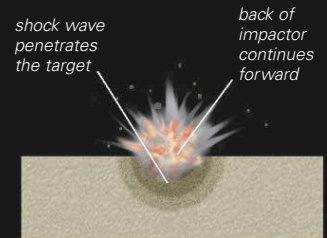


GOSSSES BLUFF CRATER

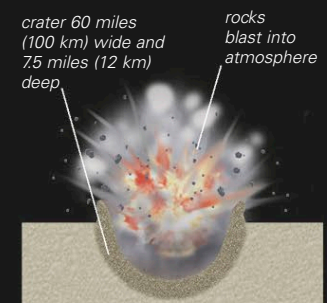
This impact crater in a central desert region of Australia resulted from an asteroid 0.6 miles (1 km) wide that smashed into Earth's surface 142 million years ago.



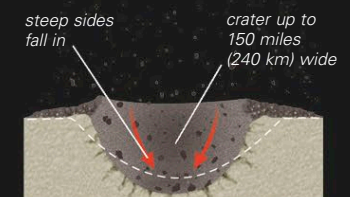
IMPACTOR STRIKES



EXPLOSION ON IMPACT



CRATER FORMATION



CRATER COLLAPSE

DEEP IMPACT

This sequence shows what typically happens when a 6-mile (10-km) wide projectile hits a rocky planet or moon. The crater formed is much larger than the impactor. The latter usually vaporizes on impact, though some melted or shattered remnants may be left at the site.

MARS

Orbits the Sun in 687 Earth days at an average distance of 141.6 million miles (227.9 million km)

VENUS

Orbits the Sun in 224.7 Earth days at an average distance of 67.2 million miles (108.2 million km)

MAIN BELT

Lies between the orbits of Mars and Jupiter and is a source of meteorites; some asteroids orbit the Sun outside the Main Belt

SATURN

Orbits the Sun in 29.46 Earth years at an average distance of 886 million miles (1.4 billion km)

NEPTUNE

Orbits the Sun in 164.8 Earth years at an average distance of 2.8 billion miles (4.5 billion km)